Bypassing Forward Models: A Comparative Study of Two Machine Learning Retrieval Prototypes for Exoplanet Atmospheric Chemistry Analysis

Introduction

Premise

Spectroscopy analysis is a brilliant technique borrowed from conventional chemistry and applied to exoplanet observation to gain insight on distant worlds' atmospheric composition.

Different molecules absorb different amounts of electromagnetic radiation. Because that effect is consistent and unique for each molecule, each of them have a distinct *fingerprint*. For example, methane (CH_4) consistently absorbs infrared light around 3.3 micrometers, leaving a sharp, predictable dip in the spectrum at that wavelength.

Observations of exoplanetary transits provide convenient conditions for deducing what atmospheric molecules are present by comparing how much light emitted from a star is dimmed during the transit, and on what wavelength and frequency.

Problem

Atmospheres are a mixture of molecules. Reconstructing the quantity and presence based on *just* the final spectroscopic reading (the 'traditional approach') has so far relied on forward modelling - which is compute-heavy. While physically accurate, this approach requires evaluating 10^4 to 10^6 simulations per planet observation [1][2]. As every new generation of satellites generates even more input data than the previous, this method is quickly becoming unfeasible as scaling effects cause diminishing returns.

The core issue is: *How do we quickly, cost-efficiently, and accurately derive meaningful estimates of exoplanet atmosphere composition based just on the spectroscopical readings?*

Novelty

Machine Learning algorithms offer an alternative solution to costly simulations. Computers can be taught to recognise patterns in an exoplanet's spectroscopical readings and, based on that, meaningfully predict that planet's atmospheric composition. Because the transmission spectrum is a one-dimensional vector, consisting of absorption values across light channels, this makes it very convenient to use as input in a ML model - the output being an estimate of the presence and amount of molecules.

This project uses the ARIEL Space Challenge 2023 data [3] to demonstrate a proof-of-concept prototype. It depicts how two different ML architectures - Random Forest and 1D CNN - tackle the same challenge, and compares their output to highlight strengths and weaknesses.

Data

The Dataset

The project utilises data from ARIEL Space Challenge 2023 [1][3]. This dataset provides realistic, simulated observations modeled after the ESA's upcoming ARIEL space telescope's technology. Per planet, the data provides four key measurements across 52 channels of infrared light:

- The actual measured transit depth (how much light was blocked).
- The exact central wavelength of that bin.
- The width of the wavelength bin.
- The simulated measurement uncertainty & noise level.

Additionally, crucial auxiliary information is provided per exoplanet, regarding said planet and its star's astronomical properties, to contextualize the observed lightwaves. Specifically:

- Stellar Information (Mass, Radius, Temperature, Distance to Star)
- Planetary Information (Mass, Radius, Orbital Period, Semi-Major Axis, Surface Gravity)

This produced a 60-feature input matrix (52 spectral channels + 8 auxiliary variables).

The training subset of the data includes seven total output properties, from which this prototype will only focus on five - those regarding the log10 abundance of five target gases - water vapor (H_2O), carbon dioxide (CO_2), carbon monoxide (CO), methane (CH_4), and ammonia (NH_3).
Because each has an upper and lower bound estimate, this produces 15 total outputs. (5 gases * 3 confidence intervals).

In total, 6766 planets were used as data, split 80/20 for training and validation.

Random Forest Regressor Model (RF)

Design

The first attempted solution to the problem was a Random Forest architecture. Because an RF is restricted to single-output targets, the model was embedded within a MultiOutputRegressor wrapper. This clones and trains 15 independent random forests simultaneously, with one dedicated to each of the 15 required chemical quartile targets.

The hyperparameter configurations were defined as follows:

- **Ensemble Size (n_estimators):** Set at a moderate *100* discrete decision trees for stable variance reduction.
- **Maximum Tree Depth (max_depth):** Set at *15* levels to prevent the trees from perfectly memorizing noise and overfitting the training dataset.

A fixed random seed (123) was enforced across the pipeline for reproducibility.

Results & Discussion

The RF model has very poor predictive value.

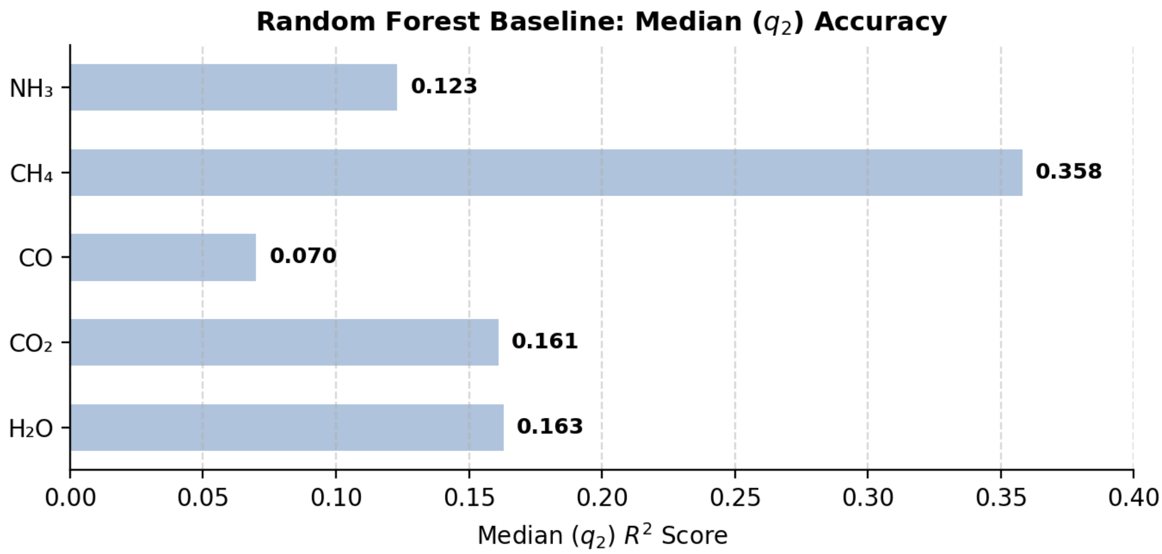


Fig. 1

Figure 1 illustrates how effectively the model mapped the input matrix to each molecular signature. We are particularly interested in the median value (q2) - its ‘best guess’ on molecule abundance. Overall, molecule explanation reaches an R^2 score of 0.358 at its best (Methane), and dips as low as 0.070 at its worst (Carbon Monoxide). Methane is the molecule that blocks the most light across a single channel, causing the sharpest dip in the spectrum; consequently, the model captures this distinct, sharp signature with the highest relative success. In contrast, carbon monoxide exhibits a fainter, more distributed absorption signature across multiple channels, which prevents the baseline architecture from effectively learning its predictive patterns.

Figure 2 performs an ‘autopsy’ on the model by studying the RF model’s feature importance. The top chart shows the 52 spectral channels by wavelength. We can see that the model managed to identify key absorption regions for H2O, CO2, and CH4, ranking their importance values around 0.05 at the highest. However, the model leans heavily on the auxiliary features instead of focusing on the actual spectrogram data. Most notably, it ranks a single planetary trait, planet_surface_gravity, higher than any single wavelength channel. This is the natural consequence of the model being tabular and treating the 52 channels in isolation - because the information derived from the channels is only meaningful when it’s more than the sum of its parts, individually, those channels end up being less useful to the model than the aux features.

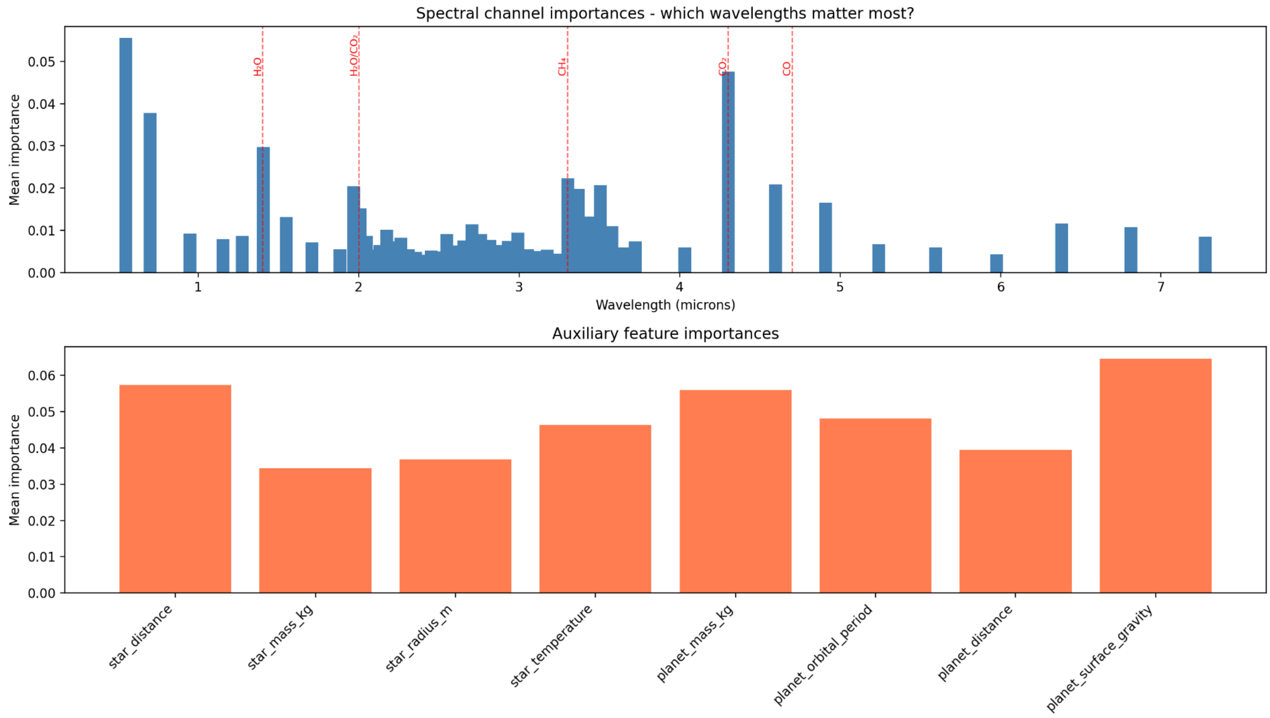


Fig. 2

In conclusion, the RF architecture has proven itself inadequate as a solution because it is “neighbor-blind” - the model does not understand that channel 3 follows channel 2, or that channel 2 follows channel 1. It looks at channels separately, and is thus unable to perceive the overall shape of the spectroscopic wave. Because of that, it must rely on cues from individual channels, causing it to miss molecules that are fainter and spread-out over multiple spectra. This is why it performs better with molecules that cause sharp dips on a single channel (CH4) and fails with molecules that cause a distributed dip over multiple channels (CO). To enable an ML model to see and study the shape of the spectrograph, we must make it neighbor-aware.

The architecture that supports this would be a 1-Dimensional Convolutional Neural Network (1D CNN).

1-Dimensional Convolutional Neural Network Model (1D CNN)

Design

For the shape of this dataset, a neural network with convolution layers provides the exact toolset necessary to analyse the overall patterns in the spectrographic slope. This architecture is a significant step up in complexity from the baseline Random Forest model, incorporating a dual-pathway feature fusion strategy. The structural design of the model is partitioned into three fundamental blocks:

Spectral Feature Extraction: The 52-channel normalized spectrum is passed as a single-input channel into a 1D convolutional stack. The first layer uses a sliding window of 5 to extract 32 low-level geometric shapes, such as local slopes and baseline shifts. The second layer applies a more granular, 3-channel wide sliding window across all 32 feature maps simultaneously, expanding channel depth to 64. This configuration forces the network to learn compound, multi-frequency absorption patterns across neighboring wavelengths and to perceive the slope as a set of recurring shape patterns associated with chemical traces. A Flatten() layer irons this multi-dimensional feature grid into a single 1D array of 3,328 extracted spectral parameters.

Aux Data Feature Fusion: The 8 static planetary aux features are not concatenated directly to the spectral stack, as this would create a significant imbalance that favors the spectral features and leads to feature dominance (as observed in early iterations of the prototype). Instead, the aux data is passed through a separate multi-layer perceptron (MLP) embedding block. This projects these static properties into a 32-dimensional latent space. This is then concatenated to the 3,328-dimensional spectral feature vector using late-fusion (torch.cat). This creates, in theory, a more balanced unified state vector of 3,360 features, allowing the global aux attributes to maintain their physical influence throughout the subsequent dense predictive layers.

Fully Connected Predictive Dense Stack: The 3,360-feature vector is fed into a sequence of linear layers that progressively condense abstract features into targeted chemical indicators. The network scales down through 128 and 64 hidden nodes using ReLU activations. The final layer maps directly to a 15-neuron output layer, corresponding to the three statistical uncertainty quartiles (q1, q2, q3) for each of the five atmospheric gases.

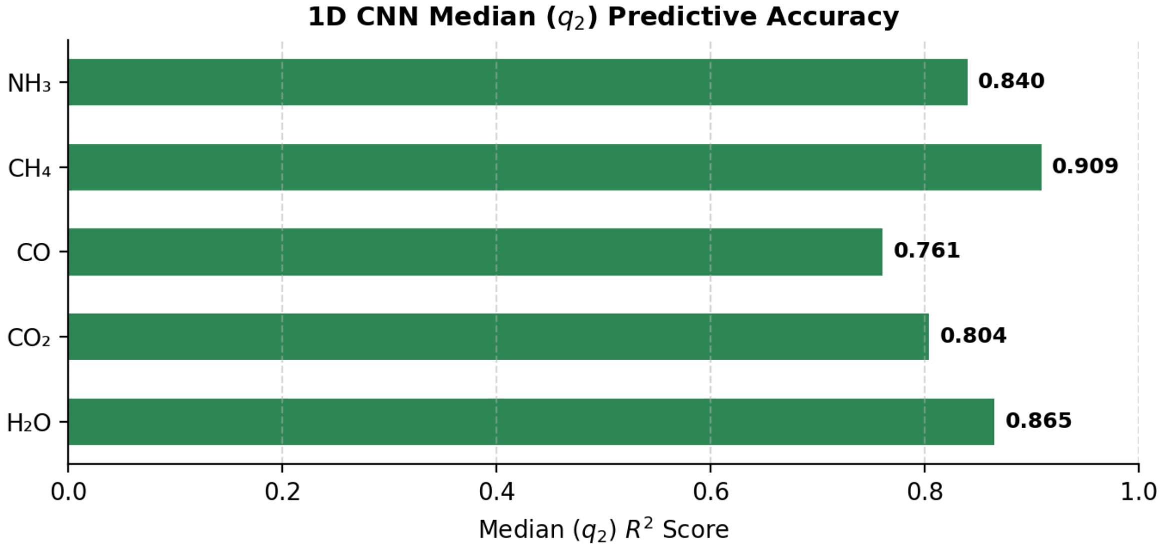
The following hyperparameters were used during training:

- Pinball Loss (Check Loss) was used because it directly optimises the model to predict specific quantiles (16th, 50th, and 84th percentiles) rather than just a mean. For each predicted value, the loss punishes underestimation and overestimation asymmetrically - pushing the output towards the desired quantile of the target distribution. This allows the network to output not just a single best guess but also a realistic uncertainty interval (q1 - q3) for every gas, without requiring a separate model or post-processing.

- Optimization Algorithm was an Adaptive Moment Estimation (Adam) with a learning rate of 10^-3

- Batching: A computationally balanced approach was chosen: 64 batches over 100 epochs.

Results & Discussion



The 1D CNN model represents a significant leap in predictive strength. It has successfully mapped spectroscopic patterns to molecular composition across all target gases, significantly outperforming the Random Forest architecture. While R^2 is inherently limited in quantile regression, it is reported here as a benchmark for the median (q2) predictive accuracy. For the most challenging target, Carbon Monoxide, this median-specific accuracy improved from an R^2 of 0.070 to 0.761. Meanwhile, at its best, it reaches R^2 = 0.909 for Methane.

This fully refutes the RF architecture, and confirms that the spectroscopic data contains complex, sequential signals that are inaccessible to tabular models but are effectively captured by the convolutional feature extraction of the 1D CNN.

Known Issues

There are several issues and limitations with this prototype that could be remedied in future versions.

The model's predictions degrade for very low gas concentrations (log abundances below approximately -8). At these trace levels, the molecular absorption signatures are near-indistinguishable from the simulated instrumental noise, causing the network to conservatively predict values near the central tendency of the training distribution rather than the true faint signal. This results in systematic overestimation of the scarcest atmospheric constituents. Potential remedies include training with a weighted loss function that up-weights low-abundance samples, or incorporating the instrument's noise profile explicitly into the input representation.

Pinball loss trains the network to target the 16th, 50th, and 84th percentiles simultaneously, but it does not impose an explicit constraint (q1 =< q2 =< q3). In rare cases, the model can output unphysical "quantile crossings" where a lower bound exceeds the median or the median exceeds the upper bound. Future versions could enforce physical realism by predicting the median and bounded offsets, or by adding a penalty term that discourages crossing.

Conclusion

This project demonstrates a proof-of-concept prototype for bypassing computationally expensive forward-modeling simulations via machine learning. By comparing two distinct algorithmic approaches on the ARIEL Space Challenge dataset, a vital structural paradigm was uncovered: tabular models like Random Forests are bottlenecked by their "neighbor-blindness," rendering them incapable of tracking continuous spectrographic slopes or resolving overlapping, faint molecular signatures like Carbon Monoxide.

By contrast, the developed 1D Convolutional Neural Network architecture was a highly effective solution. Through spatial feature extraction enabled by the convolution layers, and late-fusion of aux data, the 1D CNN successfully captured the geometric curves of the infrared spectrum, achieving a lowest median score of 0.761 and a highest of 0.909 for individual molecules.

Ultimately, this architecture proves that machine learning can decode highly complex exoplanetary atmospheres in milliseconds rather than hours. As next-generation satellites rapidly accelerate the volume of space telemetry, architectures like 1D CNN could be very useful for real-time automated exoplanetary spectrogram data analysis and a more sophisticated and refined model could see deployment and usefulness among researchers. A lightweight, quantized version of such a CNN model could even eventually run directly onboard a spacecraft.

Limitations

Currently, results were evaluated on a single 80/20 holdout split with a random seed of 123. Experimentation with other seeds demonstrated the model's R^2 dropping or increasing, meaning performance metrics are therefore dependent on the random seed used for partitioning. A more rigorous evaluation could instead average results across multiple splits and reduce sensitivity to any one particular partition.

Additionally, there are far more than five gases present in exoplanet atmospheres. Each new gas increases the complexity of the problem because of risks of spectroscopic degeneracy - similar-looking spectroscopic slopes caused by different mixtures of different input gases.

Finally, the ARIEL 2023 data is simulated to be as realistic as possible, but it remains synthetic - real data training would be preferable to prepare such a model for deployment.

References

[1] Changeat, Q. et al. (2022). "The Atmospheric Big Challenge (ABC) Database." arXiv:2206.14633

[2] Madhusudhan, N. (2019). "Exoplanetary Atmospheres: Key Insights, Challenges, and Prospects." arXiv:1904.03190

[3] ARIEL Machine Learning Data Challenge 2023. https://www.ariel-datachallenge.space/adc2023/

AI Disclosure Protocol

During the development of this project, generative artificial intelligence was used in the following ways:

- As a foundational reference point to synthesize material regarding infrared spectroscopy, exoplanetary physics, and the atmospheric retrieval process, and point to further reading resources.
- As a technical sounding board to evaluate the structural nuances, performance trade-offs, and parameter constraints of various machine learning frameworks.
- Assisted with Python syntax debugging, helper functions or line-level scripting, library documentation lookups, and formatting script visual plots.
- Aided in quickly setting up PyTorch boilerplate - mostly the standard engineering routines for the training loop, including gradient tracking, backpropagation syntax, and data batching pipelines.
- Used to quickly navigate the background documentation and structural layout of the ARIEL 2023 Challenge dataset.

Generative artificial intelligence was NOT used for:

- Writing, drafting, or phrasing any portion of the formal design report and analysis.
- Conceptualizing the project's core scope, premise, logical structure, or analytical conclusions.
- Defining the core engineering philosophy or designing the explicit configurations of the machine learning pipelines.